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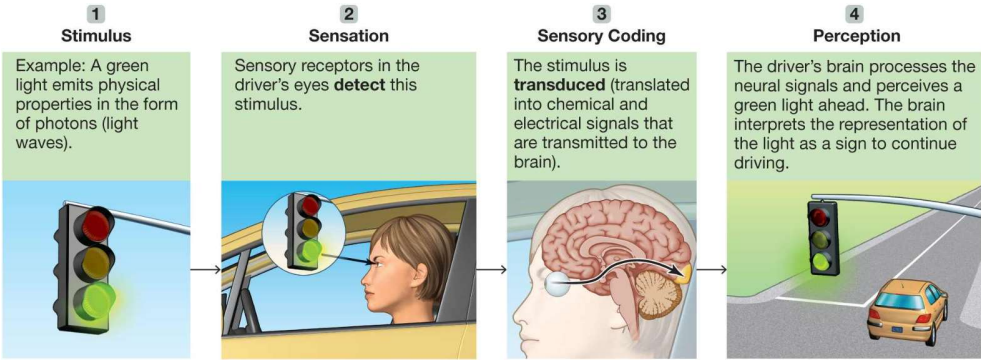
Thursday, September 12, 2024 12:34 PM

Summary of the book

How Do We Sense and Perceive the World Around Us?
5.1 Sensory Information Is Translated into Meaningful Signals



Sensation is the detection of physical stimuli in the environment. Perception is our conscious experience of those stimuli. Bottom-up processing is based on features of a stimulus. Top-down processing is based on context and expectations. Transduction is the process of converting sensory stimuli into a pattern of neural activity. Transduction takes place at sensory receptors, specialized cells within each sense organ that send signals from our environment to activate neurons. Most sensory information goes to the thalamus and then specialized brain regions.

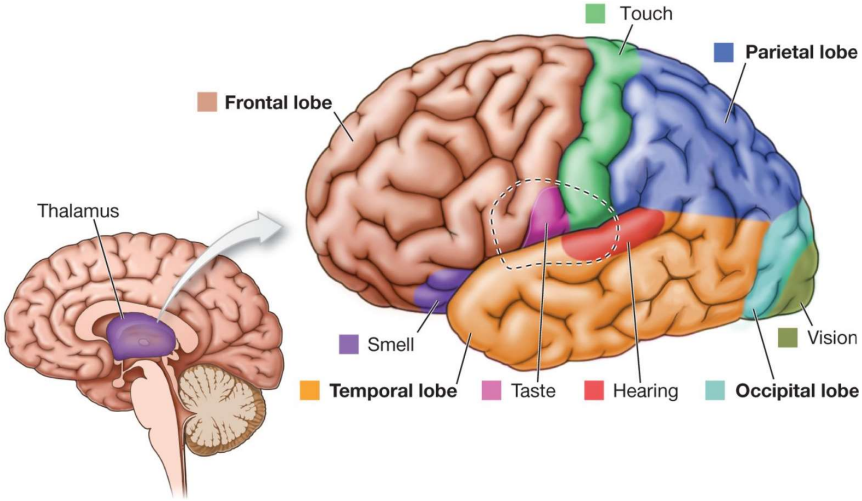
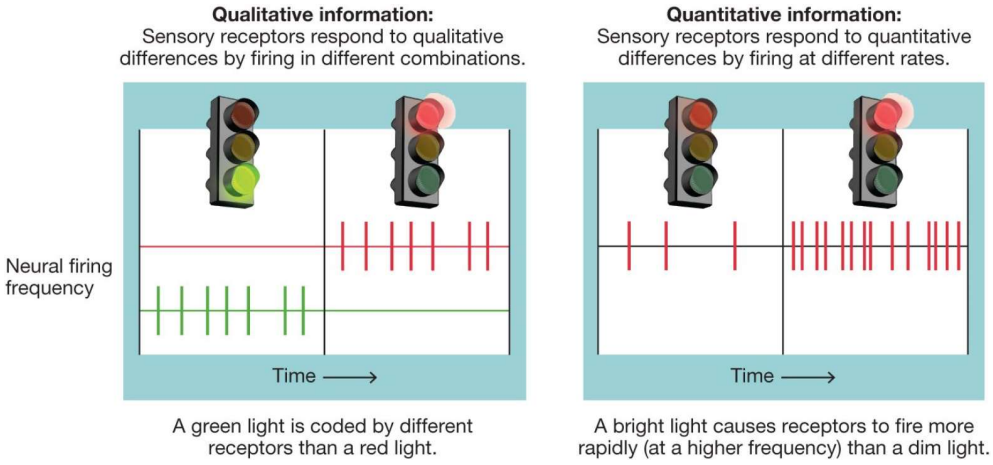


FIGURE 5.4
Primary Sensory Areas
These are the primary brain regions where information about vision, hearing, taste, smell, and touch are projected.

SENSE	STIMULI	RECEPTORS	PATHWAYS TO THE BRAIN
Vision	Light waves	Light-sensitive rods and cones in retina of eye	Optic nerve
Hearing	Sound waves	Pressure-sensitive hair cells in cochlea of inner ear	Auditory nerve
Taste	Molecules dissolved in fluid on the tongue	Cells in taste buds on the tongue	Portions of facial, glossopharyngeal, and vagus nerves
Smell	Molecules dissolved in fluid on membranes in the nose	Sensitive ends of olfactory mucous neurons in the mucous membranes	Olfactory nerve
Touch	Pressure on the skin	Sensitive ends of touch neurons in skin	Cranial nerves for touch above the neck, spinalnerves for touch elsewhere



5.2 Detection Requires a Certain Amount of the Stimulus

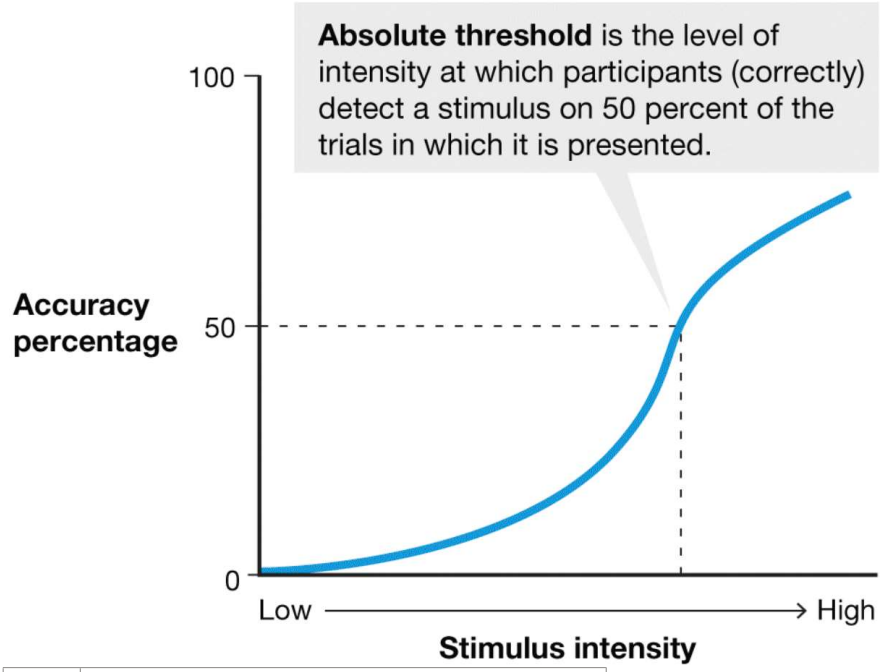


FIGURE 5.6
Absolute Threshold
This graph shows the relation between the intensity of stimulus input and a person's ability to correctly detect the input.

SENSE	MINIMUM STIMULUS
Vision	A candle flame seen at 30 miles on a dark, clear night
Hearing	The tick of a clock at 20 feet under quiet conditions
Taste	1 teaspoon of sugar in 2 gallons of water
Smell	1 drop of perfume diffused into the entire volume of six rooms
Touch	A fly's wing falling on your cheek from a distance of 0.04 inch

Information from our environment needs to surpass some level before we can detect it. An absolute threshold is the minimum amount of stimulus intensity needed to activate a sensory receptor. A difference threshold is the amount of change required for detection by a sensory receptor. Signal detection theory is about the subjective nature of detecting a stimulus. Sensory adaptation occurs when sensory receptors stop responding to an unchanging stimulus.

There are four possible outcomes when a participant is asked whether something occurred during a trial:

		Response given: Yes	Response given: No
Stimulus signal	On	Hit	Miss
	Off	False alarm	Correct rejection

Those who are biased toward reporting a signal tend to be "yea-sayers." They have many false alarms:

		Response given: Yes	Response given: No
Stimulus signal	On	89%	11%
	Off	41%	59%

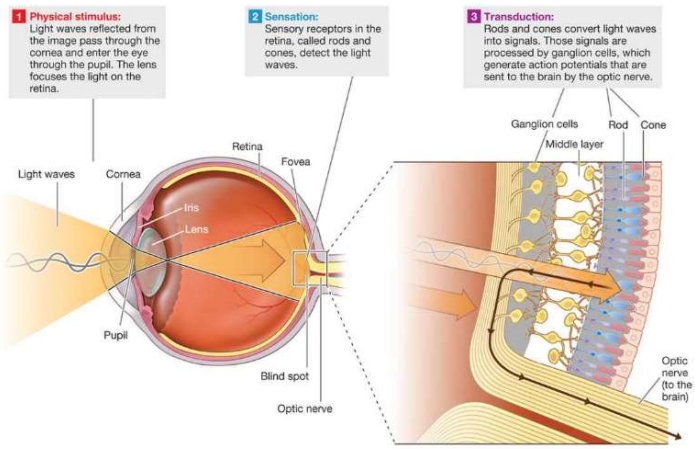
Those who are biased toward denying that a signal occurred tend to be "nay-sayers." They have many misses:

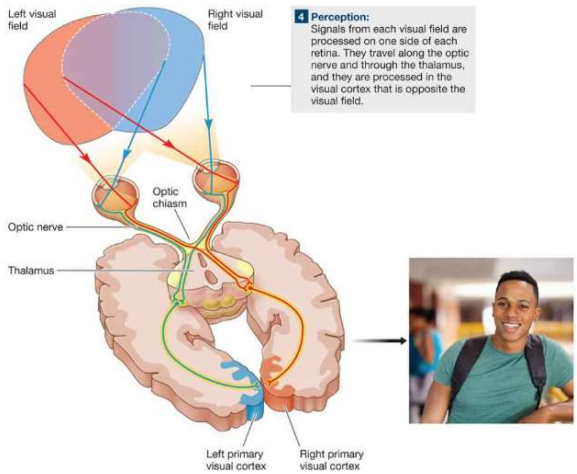
		Response given: Yes	Response given: No
Stimulus signal	On	45%	55%
	Off	26%	74%

FIGURE 5.9
Payoff Matrices for Signal Detection Theory
The percentages in this figure were invented to show representative numbers. Actual percentages vary from question to question.

5.3 The Brain Constructs a Stable Representation of the World from Five Senses
In perception, the brain integrates millions of diverse neural inputs to produce stable representations from five independent senses, which can interact at times. This activity produces awareness, a conscious experience of the physical world. In some individuals, the representations of different senses are combined in unusual ways, such as in synesthesia.

How Do We See?
5.4 Sensory Receptors in the Eye Transmit Visual Information to the Brain





Light is focused by the lens onto the retina, which is at the back of the eye. The retina houses the photoreceptors: rods and cones. Rods and cones communicate with ganglion cells of the optic nerve. This nerve exits the eye at the blind spot and crosses into the brain at the optic chiasm. There, axons from each eye cross into opposite sides of the brain, so the left hemisphere processes information from the right visual field and vice versa. The information is processed in the thalamus and the primary visual cortex (in the occipital lobe). From the visual cortex, the ventral stream processes "what" information
about objects, and the dorsal stream processes "where" information about locations.

5.5 The Color of Light Is Determined by Its Wavelength
The human eye detects electromagnetic radiation wavelengths of 400–700 nanometers. The retina contains three types of cones. Each type is responsive to a different wavelength (short, medium, or long), and this responsiveness enables us to perceive colors. Color blindness is caused by a malfunction in one or more of the cone types. Colors are differentiated by their hue, saturation, and lightness.

5.6 Perceiving Objects Requires Organization of Visual Information
Gestalt principles—such as proximity, similarity, continuity, closure, and common fate—account for the ways in which perceptual information is organized into wholes and objects. Object constancy refers to the way the brain accurately perceives images even with minimal or changing stimulus cues. The four constancies are size, shape, color, and lightness. Humans are especially good at recognizing faces. The fusiform gyrus is one of the brain regions most responsible for this ability.

5.7 You Be the Psychologist: Are Faces Special?
Psychologists debate whether faces are processed in a special way because they are important social signals or because they are objects with unique properties.

5.8 Perception of Depth, Size, and Motion Is Guided by Internal and External Cues

The visual system sees every object from two distinct vantage points:

FIGURE 5.27
Binocular Disparity

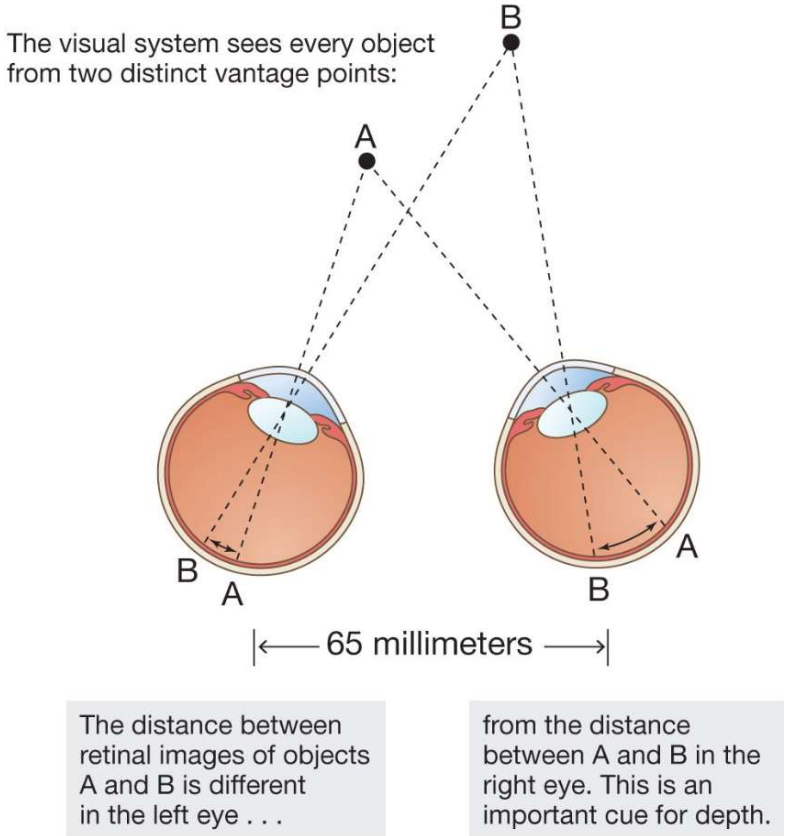


FIGURE 5.27
Binocular Disparity
To demonstrate your own binocular disparity, hold one of your index fingers out in front of your face and close first one eye and then the other. Your finger appears to move because each eye, due to its position relative to the finger, has a unique retinal image.

Depth perception is critical for locating objects in space. To perceive depth, the brain draws on binocular and monocular cues. Size perception depends on distance: Close objects produce large retinal images, whereas far objects produce small retinal images. Motion is detected by motion-sensitive neurons in the visual cortex.

How Do We Hear?
5.9 Audition Results from Changes in Air Pressure

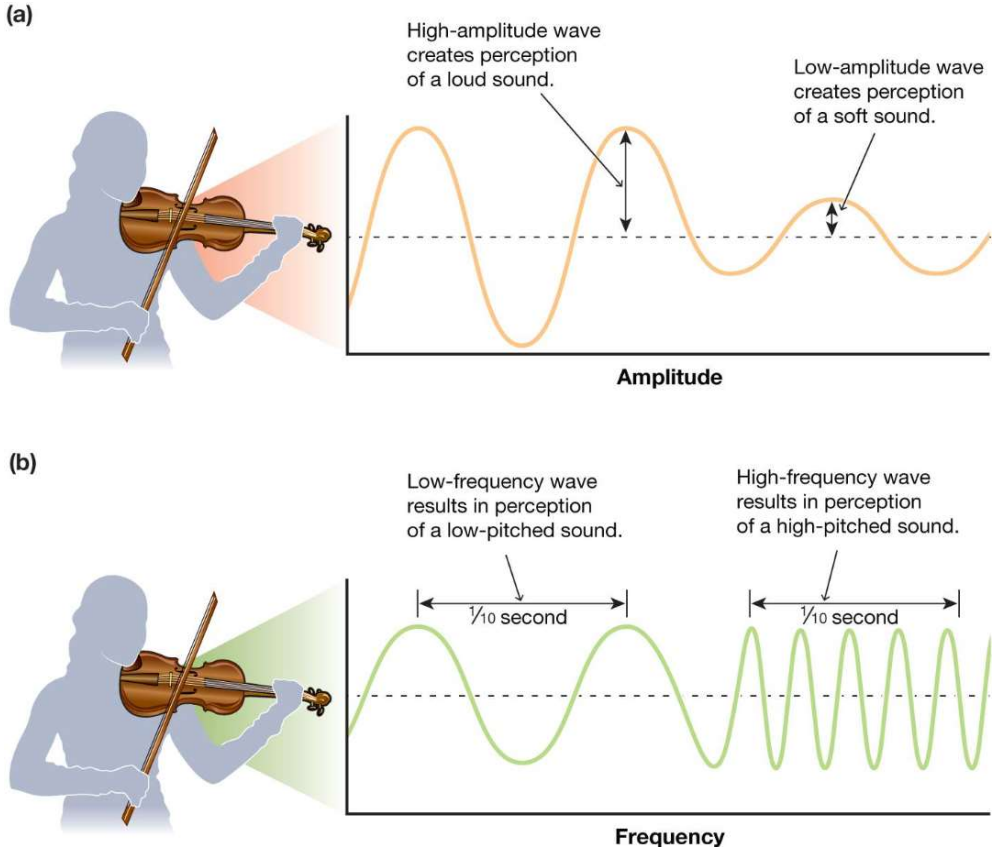
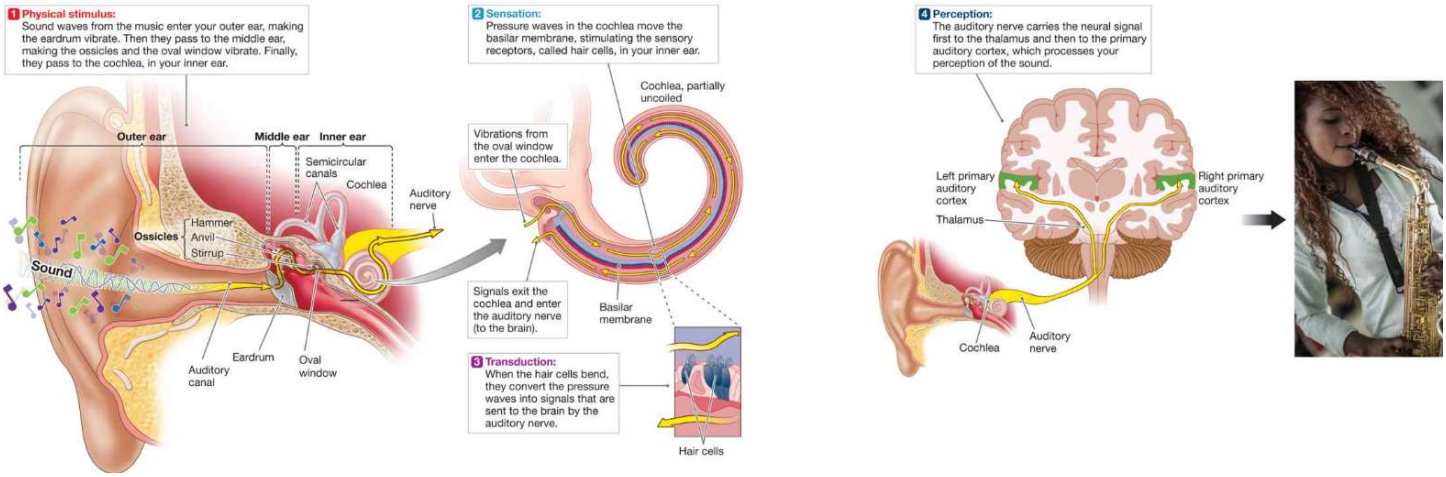


FIGURE 5.34
Sound Waves
The (a) amplitude and (b) frequency of sound waves are processed into the perceptual experiences of loudness and pitch.

The amplitude and frequency of sound waves cause the perceptual experiences of loudness and pitch, respectively. Sound waves travel from the outer ear through the auditory canal to the eardrum. Vibrations from these waves stimulate the ossicles, bones of the inner ear. The vibrations of these bones stimulate the oval window, a membrane on the cochlea, a fluid-filled chamber in the inner ear. Pressure waves from the cochlear fluid stimulate the basilar membrane. This stimulation causes the ear's sensory receptors, the hair cells, to bend and activate neurons in the auditory nerve. These neurons send messages through the thalamus to the primary auditory cortex (in the temporal lobe). Cochlear implants can help with hearing loss by directly stimulating the auditory nerve, overcoming the lack of hair cells in the inner ear.



5.10 Pitch Is Encoded by Frequency and Location

Low-frequency sound waves are sensed through temporal coding, as cochlear hair cells fire at a rate equivalent to the frequency of the waves. For high-frequency sound waves, groups of hair cells must take turns firing. High-frequency sound waves are also sensed through place coding—that is, by hair cells at different locations in the cochlea.